

DR ERNEST F. ASAMOAH

School of Biology & Environment Sciences, Queensland University of Technology



<https://github.com/efasamoah> |



asamfrt@gmail.com |



0000-0003-0712-5642

EDUCATION

Macquarie University, Australia | **PhD Environmental Science**, conferred in Feb. 2022

Beijing Normal University, China | **MSc Environmental Science**, conferred in Jun. 2017

KNUST, Ghana: **BSc (Hons) Forest Resources Technology**, conferred in Jun. 2013

PROFESSIONAL APPOINTMENTS

Queensland University of Technology, Australia | **Global Change Ecologist** (07.2024 – 06.2026)

Macquarie University, Australia | **Quantitative Spatial Ecologist & Consultant** (06.2021 – 06.2024)

The University of Queensland, Australia | **Geospatial Analyst** (11.2021 – 12.2021)

The World Bank Group | **Natural Resources & Landscape Modelling Consultant Short-Term Consultant (STC)**; 01.2022 –

ACHIEVEMENTS & INTERNATIONAL RECOGNITION

Grants & Awards

- **2024 – 2026** | QUT Postdoctoral Research Fellow (AU\$ 209,274)
- **2023 – 2024** | UNEP–Macquarie University Small-Scale Funding Grant (AU\$ 121,728).
- **2025** | NASA-MSU Professional Enhancement Award (US\$ 800) ***
- **2022 – 2023** | COVID Fellowship Scheme | Macquarie University, Australia (AU\$ 95,706).

Publications & Research Impact

- Author of 17 peer-reviewed publications, including in *Nature Climate Change* (2021), *Current Biology* (2022), and *Biological Conservation* (2026).
- Research impact: 991 citations, h-index 11, i10-index 13 (Google Scholar, April 2026).

International Collaborations & Consulting

- World Bank (2022): Contributed spatial modelling and ecosystem service analyses to Sri Lanka's **Climate Country Development Report**.
- UNEP (2023–2024): Co-developed *source-to-sea* frameworks piloted in Mozambique and Tanzania; produced policy briefs, guidelines, and training resources.
- Blue Ventures (2023–2024): Designed and delivered socio-ecological vulnerability assessments for communities in Madagascar, integrating climate, ecological, and livelihood data.

Recognition & Professional Engagement

- Featured in Macquarie University Lighthouse Magazine (2022) for research on global wilderness risks. Available through MQ's [Lighthouse Magazine](#).
- Member, Institute of Foresters of Australia (since 2018).

INVITED PRESENTATIONS IN SYMPOSIA AND CONFERENCES

- The Conference of the Ecological Society of Australia, December 2024, Melbourne, Australia

- International Congress for Conservation Biology, June 2025, Brisbane, Australia
- ESP 11 World Conference, “From global to local ecosystem services: pathways to Nature-based Solutions inspired from Down Under”, June 2025, Darwin, Australia
- China University of Geosciences (Wuhan), July 2025, Wuhan, PR China (**Invited**)

SELECTED PUBLICATIONS

*I have 17 publications in international peer-reviewed journals, including 8 first-author articles. [Google Scholar Citations](#) = 991, H-index = 11, i10-index = 13. **Author** indicates graduate advisee; **Author** indicates postdoctoral advisee. *Corresponding author.*

- Preprints -

1. **Asamoah E.F.***, Williams B., Keichihiro K., & **Rhodes RR.**, International drives unequal impacts on Nature’s Contributions to People. <https://doi.org/10.21203/rs.3.rs-8847736/v1>
2. **Maina J.**, **Asamoah E.F.***, D’Agata S, Gamoyo M, & Rakoto P. Rising Climate Risk and Loss and Damage to Coastal Subsistence-oriented Livelihoods. <https://doi.org/10.21203/rs.3.rs-3583708/v1>

- 2026 -

3. **Asamoah, E.F.***, **Mbui, J.M.**, Beaumont, L.J., Grantham, H., & Watson, J.E.M. (2026) Global forests risks to climate and land-use change. *Biological Conservation*.

- 2024 -

4. Twumasi-Ankrah M.J., Zhan J., & **Asamoah E.F.** (2024). Mapping ecoregional vulnerability to climate change for Africa. *Science of The Total Environment*, 953, 176219.
5. Chanachai J., **Asamoah E.F.**, **Maina J.M.**, Wilson P.D., Nipperess D.A., Esperon-Rodriguez M. & Beaumont L.J. (2024). What remains to be discovered: A global assessment of tree species inventory completeness. *Diversity and Distributions*, e13862.

- 2022 -

6. **Asamoah E.F.*** & **Maina J.M.** (2022). Nature-based climate solutions require a mix of socioeconomic and governance attributes. *CellPress: Iscience*, 25(12), 105699.
7. **Asamoah E.F.***, Di Marco M., Watson J.E.M., Beaumont L., Venter O., & **Maina J.M.** (2022). Land use and climate risk assessment for Earth’s remaining wilderness. *CellPress: Current Biology*, 32 (22), 4890–4899.e4.
8. Halefom T.H., Pullar D., Pojani D. & **Asamoah E.F.** (2022). How much traffic stress can cyclists endure? *Case Studies on Transport Policy*, 10(4), 2251–2261.
9. **Asamoah E.F.*** & **Maina J.M.** (2022). Hunger needs to be the priority for biomass (In VOICES – “The Limits of Biomass”). In *One Earth*, 5 (1), 3–6.

- 2021 -

10. **Asamoah E.F.***, Beaumont L., **Maina J.M.** (2021) Climate and land-use changes reduce the benefits of protected areas. *Nature Climate Change*, 11, 1105–1110.
11. Destek M.A., Sarkodie S.A., **Asamoah E.F.** (2021). Does biomass energy drive environmental sustainability? An SDG perspective for top five biomass consuming countries. *Biomass and Bioenergy*, 149, 106076.

- 2019 -

12. Arthur M., Liu G., Hao Y., Zhang L., Liang S., **Asamoah E.F.**, Lombardi G.V. (2019). Urban food-energy-water nexus indicators: A review. *Resources, Conservation and Recycling*, 151, 104481.
13. Rutebuka E., Olorunnisola A.O., Taiwo O.J., Mwaru F., **Asamoah E.F.**, Rukundo E. (2019). Quantitative review of ecosystem services and disservices studies in the tropics. *Open Journal of Ecology*, 9, 85–106.
14. Sarkodie S.A., Strezov V., Weldekidan H., **Asamoah E.F.**, Owusu P.A., Doyi I.N.Y. (2019). Environmental sustainability assessment using dynamic Autoregressive-Distributed Lag simulations—Nexus between greenhouse gas emissions, biomass energy, food and economic growth. *Science of The Total Environment* 668, 318–332.

- 2018 -

15. Rutebuka E., Zhang L., **Asamoah E.F.**, Pang M. & Rukundo E. (2018). Emergy analysis of resource dynamics of Rwandan Economy. *Sustainability*, 10, 1791.
16. East Africa. Rukundo E., Liu S., Dong Y., Rutebuka E., **Asamoah E.F.**, Xu J., & Wu X. (2018). Spatio-temporal dynamics of critical ecosystem services in response to agricultural expansion in Rwanda. *Ecological Indicators* 89: 696–705.

- 2017 -

17. **Asamoah E.F.***, Zhang L., Liang S., Pang S., Tang S. (2017). Emergy perspectives on the environmental performance and sustainability of small-scale gold mining in Ghana. *Sustainability*, 9 (11), 2034.
18. **Asamoah E.F.***, Zhang L., Liu G., Owusu-Prempeh N., Rukundo E. (2017). Using historical image data to estimate the “forgone” ecosystem service value for small-scale gold mining. *Sustainability*, 9 (11), 1976.

- 2014 -

19. Siaw DEKA, **Asamoah E.F.**, Gyabaa B. (2014). The stocking and socio-economic uses of *Borassus aethiopum* in Abrimasu Forest Reserve of the Mampong Forest District of Ghana. *Journal of Ene. And Nat. Res. Man.*, 1 (3), 148–155.

Teaching & Tutorials

2025 | Case Studies in Environmental Science | **Queensland University of Technology**

2023 | Introduction to GIS and Remote Sensing | **Macquarie University**

2014 | Introduction to Geographic Information Systems | **KNUST**

2014 | Ergonomics and work study | **KNUST**

MSc Mentoring

Jariya Chanachai (2022). What remains to be discovered: A global assessment of tree inventory completeness. Awarded MSc with Higher Distinction

Current projects

- Climate connectivity can inform cost-effective ecosystem-based mitigation for enhanced biodiversity co-benefits. MQU (lead)
- Global dynamics of land sharing and land sparing. QUT (lead)

REFERENCES

Names	Institutions and position	Address
Prof. Jonathan Rhodes (Supervisor)	Capacity Building Professor, Queensland University of Technology, Australia	jon.rhodes@qut.edu.au Tel: 3138 5023
A/Prof. Joseph Maina (PhD supervisor & Employer)	Senior lecturer in Geo-Information Systems, Macquarie University, Australia.	joseph.mbui@mq.edu.au Tel: +61 2 9850 9448
Prof. Linda Beaumont (PhD supervisor & Employer)	Deputy Dean, Education and Employability, Macquarie University, Australia.	linda.beaumont@mq.edu.au Tel: +61 2 9850 8157

I certify that the information above mentioned is true, complete, and correct to the best of my knowledge and belief.

Ernest Frimpong Asamoah, PhD

Saturday, May 30, 2026